On PIE Full Grades in Some Zero-Grade Contexts: **-tí-, *-tó-*

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1. Background.

1.1. Despite the fact that primary *ti*-stems (nomina actionis, abstracta) generally appear with zero grade of the root (Ved. *matí-* 'thought, devotion', etc.), it seems to be all but universally assumed that such formations originally inflected according to the proterokinetic pattern, with full-grade root and zero grade of the suffix in strong forms, and zero-grade root with full grade of the suffix in weak forms.[1] The matter has recently been discussed in a particularly clear and useful way by M. MEIER-BRÜGGER (2000: 194f.).

Proterokinetic inflection as such, however, is not directly attested for *ti*-stems. Thus, it is normally assumed that zero grade of the root, the oxytone accentuation characteristic of the weak stem, and the zero-grade suffix of the strong stem have all been generalized, thus yielding the typical einzelsprachlich appearance of primary *ti*-stems — i.e., a generalized "R(Ø)-tí-" (as in Ved. *matí-* etc.).

1.2. Two types of more or less direct evidence are usually cited in favor of these assumptions. For the root syllable, we have accented zero grades of the type Ved. *gáti-* 'moving', *śrúti-* 'hearing', etc.,[2] which are thought to point in the direction of the original accented full grade. For the suffix, we have several bits of evidence (all recently discussed by MEIER-BRÜGGER, loc. cit.), pointing to a full-grade suffix in weak cases and (less directly) to a full-grade root in strong cases:

(i) Vedic gen. sg. *-tés*

(ii) Gk. (Att.) gen. sg. *-σεως* (with quantitative metathesis; *-ē-* perhaps from loc.sg. **-ēi*), dat.sg. *-σει*

[1] In the formalism used here for convenience: strong R(é)-ti- ~ weak R(Ø)-téi̯- ("R" = root).

[2] Listed in LUBOTSKY (1988: 34f.).

(iii) Ved. acc. Pl. *matís* (implying, in its paradigmatic context, **mán-ti-ns*)

1.3. At least according to the older handbooks, still more direct traces of full-grade/zero-grade ablaut have been seen in a few parallel forms of the type Att. *δέρρις* 'skin, hide' : *δάρσις* 'tearing, separation' (Gal.), Ion. [Hdt.] *ἄγερσις* 'assembly' : W.Ion. *ἄγαρρις*, cf. Arc. *πανάγορσις*, Hsch. *ἄγορρις* (e.g. SCHWYZER 1939: 505, following PEDERSEN). But none of these alleged relics is compelling: most of them are better explained in other ways,[3] especially given the documented tendency of Greek to develop postverbal full grades for original zero grades, of the type *φεύξις* 'flight' (Soph.) ← *φεύγω* 'flee' (vs. Hom. *φύξις*).[4] In a similar way, some other full-grade *ti*-stem forms have long been suspected of having arisen secondarily, as with some Indic forms found beside parallel full-grade *tu*-stems (see e.g. LIEBERT 1949: 50-1 on Ved. *tantí-* 'cord' : *tántu-* 'thread').

1.4. Given, then, the relatively sparse direct evidence for the proterokinetic inflection of *ti*-stems, together with the nearly canonic zero grade of the root that appears throughout Indo-European, it is not surprising that the generalization of the zero-grade root in the strong forms is often thought to have occurred quite early, as in H. RIX's formulation: "Vollstufe der Wurzel (Nom.Akk.) ist wohl schon spät- idg. durch Schwundstufe ersetzt" (1992: 146). In a more extreme way, some scholars have gone so far as to deny the existence of any evidence at all for a mobile inflection: thus, according to A. SIHLER, "there is no actual evidence for tonic kinesis" (1995: 279). But it seems to be largely unremarked, in such discussions, that there is a class of full-grade *ti*-stems — or actually, two related or overlapping subclasses of such forms — that is in fact rather well-attested. I refer, to begin with, to the first members of *τερψίμβροτος* compounds, where "CéH-ti-" forms (of the type RV *dā́tivāra-* 'freigebig') are prominent.

This material calls for two important qualifications at the outset, the first of which concerns the nature of the first members in question. While it is generally agreed that the core of this formation does indeed

[3] Thus *δέρρις* may not in fact reflect **δέρσις*, but could show expressive gemination (see FRISK 1.369, s.v. *δέρω*, with further references).

[4] Hence, in fact, *ἄγερσις*, easily secondary to *ἀγείρω* 'gather, collect' (cf. CHANTRAINE 1933: 280, including *ἔγερσις* ← *ἐγείρω*, among other such forms).

involve primary *ti*-stems, G. DUNKEL has argued (in my view rather persuasively) for a "polygenetic" origin (DUNKEL 1992), whereby several other types of formation besides primary *ti*-stems have contributed to the body of so-called τερψίμβροτος compounds. But this polygenesis should not be a factor here, since the τερψίμβροτος first members to be treated below are in general not to be accounted for as adverbial forms, imperatives, or, in short, anything other than *ti*-stems. The second qualification involves the standard assumption (see e.g. SCHINDLER 1997: 539) that many of the full grades found as τερψίμβροτος first members are secondary in one way or another, typically related to (synchronically) full-grade verbal stems (such as *s*-aorists). Dunkel has argued (again persuasively, in my view) that Greek forms of this sort are in general sigmatic. Thus, since I will mainly refer to Greek forms with unassibilated *-t-*, the potential complicating factor of secondary full grades should likewise not enter the picture in any significant way. My initial claim will be, quite simply, that among τερψίμβροτος first members that have the appearance of continuing old *ti*-stems, full-grade forms built to laryngeal-final roots of the shape "CeH-" have a certain prominence, cf. the Vedic Musterbeispiel *dā́tivāra-* cited above, whose first member has the same structure as a Homeric form like βωτιάνειρα 'nourishing men' (A 155) < *g^weh_3-ti-*.[5]

There exist, moreover, at least a few archaic-looking simplex forms of exactly the same type, such as the *ti*-stem **méh$_1$-ti-* (: **meh$_1$-* 'measure') that underlies Hom. μῆτις and other forms (RV *-māti-*, Lat. *mēti[or]*, OE *mæþ*). I will suggest, then, that there may be something

[5] The idea that this form, like ὠλεσίκαρποι 'shedding fruit' (κ 510), shows an "Aeolic" reflex of metrical lengthening (i.e. βωτι˙, ὠλεσι˙ for expected **βουτι˙, **οὐλεσι˙; thus HOUSEHOLDER and NAGY 1972: 68, apparently followed by JANKO 1992: 17n30), has little to recommend it. As seen by SCHULZE, SOLMSEN and others, there are a number of secondary sources for the ὠ- of ὠλεσίκαρποι (see WYATT 1969: 61-3); indeed, one such source or contributing factor might be the more archaic type of τερψίμβροτος compound seen in βωτιάνειρα itself (vs. βόσιν T 268), which, unlike ὠλεσίκαρποι, has an identifiable historical source for the -ω- — namely, the type Ved. *dā́tivāra-*, and full-grade simplex "CéH-ti-" forms more generally (see the immediately following text).

systematic in the relatively clear-cut survival of full-grade *ti*-stems of exactly this sort.[6]

1.5. There is nothing surprising or problematic as such about a correlation between root structure and a particular morphological category — a feature otherwise familiar, for example, in the Indo- European root noun, as highlighted by SCHINDLER (1972a, 1972b) and many others since. Such potential correlations, however, deserve to be verified, investigated, and (if possible) accounted for. In the present case, the issue is also to be viewed in connection with the broader question of how to account for the special nature of *ti*-stem inflection as we actually find it (§§1.1.-1.4.), i.e., with relatively little evidence for full-grade roots apart from several phonotactically restricted subsets, such as roots ending in a laryngeal.

To conclude the introductory exposition, I suggest further that this pattern (i.e., "late IE" CéH-ti- ~ R(Ø)-tí-) may not be entirely isolated, since similar behavior can be found among verbal adjectives in **-tó-* (i.e., CeH-to- ~ R(Ø)-tó-). On the face of it, such a comparison may seem surprising: an oxytone thematic (and therefore non-ablauting) formation with canonic zero-grade of the root is after all quite different from an athematic formation that is assumed to have been ablauting, at least at some stage of the protolanguage. But in terms of the late Indo-European descriptive profile of *ti*-stems, we do in effect have a nearly canonic oxytone formation with almost pervasive zero grade of the root, and this is the sense in which I propose to examine the comparison with formations in **-tó-*.

2. Full-grade **-tó-* (especially C(R)eH-tó-)

2.1. In considering full-grade verbal adjectives in **-tó-*, it should be clear that certain types of forms are not of interest. We can exclude, for example, familiar obstruent-cluster substitutions TT-tó- → TeT-tó- (**pek^w-tó-*, **seĝ^h-tó-*, etc.). Also excluded here are various substantivizations with secondary full grade, such as result nouns with *e*-grade

[6] I cannot agree with RUIJGH (1997: 267) that μῆτις is actually the phonologically regular result of a zero-grade **mh$_1$-ti-*. But even if this development should turn out to be correct for Greek, there can be little doubt about the existence of a full-grade **méh$_1$-ti-* (cf Lat. *mētior* etc.).

(*$\hat{k}l\acute{e}u$-to- 'sound' [Av. *sraota-*], etc.) or *o*-grade (*νόστος* 'return', etc.).

2.2. What remains is a rather well-attested body of full- grade verbal adjectives with root-final laryngeal, although the pattern does not seem to have attracted much notice. The limited attention accorded such material has mainly been directed to those forms with resonant or glide after the initial stop (i.e., "CReH-to-" forms): in general, these have traditionally been taken to reflect secondary (and relatively late, indeed mostly einzelsprachlich) creations, no doubt correctly in many instances.[7] Thus, despite some apparent equations (Lat. *nētus* : Hom. *ἐΰννητος*, Lat. *-pōtus*, *nōtus* : Ved. *prātá*, *jñātá-*, etc.), a series of such forms in Latin can be satisfactorily accounted for on the basis of full grades elsewhere in the system, typically in long-vowel perfectum forms that go back to active root aorists: Lat. *nētus*, *-plētus*, *nōtus*, *crētus*, *sprētus* etc. (for phonologically regular ***nātus*, ***-plātus* etc.) after *nēvī*, *-plēvī*, *nōscō*/*nōvī*, etc. (cf. SCHRIJVER 1991: 340f.).

2.3. It has occasionally been suggested that a few forms of this sort may be relatively old, as with F. O. LINDEMAN's claim (1997: 98ff.) that a number of these items should be reconstructed for "some stage of the protolanguage" (citing apparent equations like Lat. *-plētus*: Ved. *prātá-* : [?] Arm. *li* and *nōtus* : *jñātá-* : *γνωτός*). While I am not in sympathy with the use to which LINDEMAN has put such forms,[8] it still seems legitimate to consider, on its own terms, the question of whether some of them may indeed be inherited.

One interesting group involves forms that correlate with a 4th-class or *i̯e/o*-present, where full grade for laryngeal-final roots was apparently normal; e.g.:

(i) *$pleh_3$- 'swim, flow' (LIV 437: aor. *ἀπέπλω*) *πλώω* : *πλωτός*.

[7] See in general WACKERNAGEL-DEBRUNNER II.2: 565ff., KURYŁOWICZ 1956: 251f.

[8] LINDEMAN adduces them as ammunition against the triple representation of laryngeals in Greek: if a form like Lat. *-plētus* beside Ved. *prātá-* is an old state II full grade, then (in LINDEMAN's view) a form like *-γνητός* (in *κασίγνητος*) could equally well be a state II full grade and not a zero grade, and similarly for alleged zero-grade forms built to roots ending in the other two laryngeals.

(ii) *$\hat{d}^{h}meH$- (beside *$\hat{d}^{h}emH$-) 'blow' — *dhmāya-* (Up.+ but perhaps old): RV *dhmātá-* (RV pres. *dhámati*)

(iii) *$mleh_2$- 'become soft etc.' — Ved. *mlāya-* (AV)[9] : RV *mlātá-*, Av. *(a)mrāta-*

(iv) *$\hat{k}reH$- 'be cooked, boil' — RV *śrāya-* : *śrātá-*

Thus, even if the CReH-tó- forms are ultimately derived secondarily from full-grade presents in CRéH-i̯e-, this process may have taken place relatively early in at least a few cases, although there is no space here to explore various complications posed by many of these forms.[10]

2.4. We also find some relatively isolated material, including items lacking parallel full grades that might have served as the basis for the full-grade verbal adjective. Thus:

(i) beside Ved. *ghrātá-* (B., also *ghrāṇa-*), the reduplicated present *jíghrati* (RV, AV) is older than the root present *ghrāti* (KB+), and *ghrātá-* itself (*$g^{wh}reh_1$-tó-) may have at least one attractive cognate elsewhere (the *-φρητός* of *ὀσφρητός*), and possibly two (cf. E. *breath*; e.g. WATKINS 2000: 35, s.v. **g^{wh}hrē-**); similarly:

(ii) *$u̯reh_1$- 'find': OIr. *fríth*

(iii) *$k^{w}i̯eh_1$- 'rest': Lat. *quiētus*, GAv. *šiiāta-*/OP *šiyāta-*

(iv) various (partly ambiguous, to be sure) Greek items, especially onomastic forms: e.g. *˙πᾱστος (*$\hat{k}u̯eh_2$-to-*, with secondary *-σ-*; *πέπᾱμαι* 'possess'), *˙κτᾱτός (*$tkeh_2$-to-*, cf. LIV 562; *κτάομαι* 'obtain'), *˙μνᾱτός (*$mneh_2$-to-*? cf. Skt. *mnāta-* [S.]; *μνάομαι* 'woo').

2.5. Full-grade CeH-to- forms — i.e., full-grade verbal adjectives to laryngeal-final roots without an initial glide or resonant cluster — seem to be rather rare, though they are attested with some certainty in Indo-Iranian. Again, the limited notice taken of such forms has focused

[9] A *$mleh_2$-i̯e-* perhaps also in Lat. *blandus* 'smooth, alluring' (MEISER 1998:112).

[10] I note in passing that (as I hope to discuss elsewhere in more detail) the original *$\hat{k}r[H]$-tó- to *$\hat{k}reH$- 'be cooked' could conceivably underlie Ved. *śṛtá-* 'cooked, boiled', if this is ultimately a decompositional form (cf. RV *śṛta-pā́-*, TB *śṛta-kāma-*, etc.) with laryageal loss of the type Ved. *grumuṣṭí-* (differently J. NARTEN [1987: 273], via anit *$\hat{k}er$- vs. enlarged *$\hat{k}r$-eH-).

on their supposed secondary status. Thus Lat. *fātus* is said to have a "generalized full grade" (SCHRIJVER 1991: 402), after the present stem of *for/fārī* 'speak'. The recent status of some other forms seems more questionable. What, for example, is the real basis for considering the isolated Lat. *pōtus* 'having drunk, having been drunk' to be a recent creation, as is generally assumed (e.g. SCHRIJVER 1991: 341)? Admittedly, the situation with this root is complex, given an overt "long-diphthong" form in Ved. *pītá-* beside the ordinary zero grade in Gk. ποτός. For *pōtus*, however, we have in Italic only the old reduplicated present **pí-ph$_3$-e/o-* and an instrument noun **peh$_3$-tlo-* (disregarding Lat. *pōtiō*, which can simply be based on *pōtus*). To be sure, in Indo-European perspective this verb had a full-grade active root aorist (Ved. *ápāt* etc.), among other formations with full grade (see LIV 417); but no trace of such forms survives in Italic. Similarly, Lat. *fētus* (adj.) 'pregnant, teeming' (a form not treated in SCHRIJVER 1991) could continue an old **dʰeh$_1$-to-* (: **dʰeh$_1$(i)-* 'suckle', cf. Ved. *dhītá-*), though in principle a Lat. **fēto-* could have been formed after a present **fē(i)ō* (> Class. **feō*),[11] cf. the old middle participle in *fēmina* 'woman'.

2.6. One of the most attractive cases for a potentially old CeH-to- form has recently been discussed by J. L. GARCÍA RAMÓN, in connection with Gk. ζητέω 'seek, ask' (1999a: 88). According to GARCÍA RAMÓN, ζητέω is based on a verbal adjective **ζᾱτό-*, which in turn is directly equatable with the Indo-Iranian **yātá-* seen in Ved. *˚yāta-*[12] and Av. *yāta-*, to a root **i̯eh$_2$-* 'eifrig suchen, fordern'.[13] GARCÍA RAMÓN seeks to support his reconstruction by appealing to the existence of

[11] Probably *not* reflected (indirectly) in the post-Augustan verb *fētāre* 'impregnate', which is more likely a denominative (with factitive meaning) based on *fētus* than an iterative-intensive formed directly to the old **feō*, if such a verb existed.

[12] RV *ávayātahe*ḷ*as-* (I.171.6b), cf. RV *yātár-*, *avayātár-*, *ṛ̯nayá-* (GARCÍA RAMÓN 1999a: 81ff.).

[13] The traditional reconstruction **di̯eh$_2$-* for the root shape (maintained in LIV 107) precludes aligning the Greek material with that from Indo-Iranian (and perhaps Tocharian, see LIV ad loc., fn. 1). Otherwise, of course, proponents of the "Lex Francis-Normier", according to which a zero grade **dih$_2$-to-* would regularly yield Pr.-Gk. **di̯āto-* (and not **dīto-*), hold this up as one of the prime examples of this treatment (see RASMUSSEN 1990/91[1999] for general discussion, and for this analysis of **ζᾱτό-*, see most recently KAZANSKIJ 2000: 34).

similar full-grade verbal adjectives in Indo-Iranian (Ved. *snātá-* and others, more on which below), with the clear implication that what underlies the Greek/Indo-Iranian equation is an inherited full- grade verbal adjective **i̯eh$_2$-tó-*. One should note, incidentally, that the Greek verbal adjective may not need to be reconstructed, given the actual existence of ζᾱτός* 'sought for' in Arcadian (IG V:2.4.22 = SCHWYZER Del. 654g.22, ζατον; Tegea, 4th c.), although GARCÍA RAMÓN does not mention this form. The interest of this (*)ζᾱτό- increases if we consider that it may be one of the very few forms of its type in Greek that has the appearance of being inherited.[14]

2.7. As just noted, GARCÍA RAMÓN cites several forms of the sort Ved. *snātá-* in support of his reconstructed **i̯eh$_2$-tó-*. But, even apart from certain minor details concerning those forms,[15] the Indo- Iranian situation with regard to such patterns is more complex. In Iranian, full-grade forms of the sort in question have actually been generalized — thus Av. *dāta-/-dāta-* (*dā-* 'setzen'), *rāta-* (*rā-* 'give'), *māta-* (*mā-* 'measure'), likewise OP *dāta-/-dāta-*, *-māta-*, *pāta-* (*pā-* protect')[16] — a factor which, while supporting the existence and even the antiquity of the type, weakens the comparative value of the Iranian side of particular equations like Ved. *˚yāta-* ~ Av. *yāta-* (cf. §2.6.). Such a generalization has clearly *not* taken place in Indic, as the following survey of Vedic material shows:

(i) TeH- (including seH-, sTeH-):	*Titá-*	*dita-* (*dā-*'bind')
		hitá-/-dhita- (*dhā-* 'put')
		sitá-/-sita- (*sā-* 'bind')
		-sthita- (*sthā-* 'stand')
		śitá-/-śita- (*śā-* 'sharpen')
	cf.	*jahitá-* (*hā-* 'leave')

[14] In theory, a similar form (*τᾱτό-, cf. τήτη˙ ἀπορία, ἔνδεια, στέρησις Hsch.) may underlie τητάομαι 'be deprived of' [Hes. *Op.* 408+], although this is certainly not the only possible derivational source for this verb, and there is in any case no direct external support for a full-grade verbal adjective to **(s)teh$_2$(i)-* 'steal'.

[15] GARCÍA RAMÓN's "ved. *dhātá-*" does not seem to exist, and despite his "**bhātá-*", such a form does not occur with accent (*bhāta-*, *˚bhāta-* Up.+).

[16] See HOFFMANN-FORSSMAN 1996: 245, KENT 1953: 78, KURYŁOWICZ 1956: 249f.

(ii) ReH-: *Rātá-* — *yātá-* (*yā-* 'go'), *-yāta-* (*yā-* 'eifrig suchen', §2.6.), *rātá-* (*rā-* 'give')[17]

Although the numbers are not large, the distributional pattern for CeH- roots appears to be complementary: if the first consonant of the root is a non-sonorant obstruent or an obstruent cluster, the result is the regular and predicted zero grade of the root, with vocalized laryngeal, as in *-dita-*, *hitá-*, *sitá-* etc. (including *jahitá-*, with secondary reduplication); but if the first consonant is a sonorant, the full grade appears, as in both versions of *yātá-* and in *rātá-*.[18]

There are indications of similar behavior elsewhere, although these deserve more extended treatment than can be provided here, and much of the material is in some respects equivocal.[19] The essential point for now is the suggestion that the Vedic pattern in (i) vs. (ii) above may be relatively old, such that full-grade "substitutes" like Ved. *yātá-*/*-yāta-*, *rātá-*, Av. *yāta-*, *rāta-*, and OP *māta-* reflect the earlier core of a process that ultimately results in generalized full grades for forms with intial stop such as Av./OP *dāta-*, OP *pāta-*, and late Ved.

[17] On the original sense of this root in IE see now MELCHERT 1999a.

[18] In the Rig-Veda, the only two exceptions to this pattern appear to be *tvā́dāta-* (modeled after *tvā́datta-*, which has a reduplicated form similar to *jahitá-*, except with laryngeal loss), and *-mita-* (*mā-* 'measure'), which occurs once in a patently late hymn (I.164.45a) and a second time in a hymn that may not be much older (VI.7.6a). For the secondary status of *-mita-* (*párimitā* I.164.45a, *vímitāni* VI.7.6a), it may also be significant that in the Rig-Veda, *Rātá-* forms never occur with preverb, while the *Titá-* forms are attested much more commonly with preverb than without (approx. 2:1). Insofar as compounded **-tó-* is an archaic feature (MEILLET 1929), this seems to confirm the relatively late character of the *Rātá-* forms.

[19] In Gmc., e.g. OE *mǣþ* n. 'mowing' < **[h₂]méh₁-to-* (with accent retraction, perhaps associated with the substantivization) contrasts with ordinary zero-grade forms like OE *sæd*, Go. *sad*, OHG *sat* satisfied' < **sh₂-tó-*. In Latin, note the problematic status of forms like *ratus* (: *rērī* 'think'), **lato-* (cf. *latēre* 'lie hidden') and **nato-* (cf. *natāre* 'swim'), which could be analogical substitutes ("morphological zero grades") modeled on normal zero grades like *datus*, *status*, *satus*, **fato-* (: *fatērī* 'assert') etc. (cf. Ved. *mita-* above), although such forms are also considered by some simply to reflect the regular phonological development of inherited zero grades (see e.g. SCHRIJVER 1991: 171, and B. IRSLINGER [this volume]).

tvā́dāta-, *-bhāta-* etc.[20] This is not, of course, an entirely new conception: it corresponds, in essential points, with KURYŁOWICZ' classic analysis of the Indo-Iranian material (1956: 251f.), in terms of the rise of a generalized *(T)Rā(T)* in Indic (especially *(T)Rā-: (T)Rā-ta-)*, replacing *(T)Raə(T): (T)R̥əT)*, as part of a general replacement of samprasāraṇa ablaut. The novelty here is that in view of GARCÍA RAMÓN's equation of Indo-Iranian **yātá-* with Gk. (*)ζᾱτό-, we have been led to consider whether the substitution of a full grade for some types of samprasāraṇa zero grade may belong to some stage of the protolanguage.[21]

3. Full-grade **-tí-* (esp. CéH-ti-)

3.1. As noted above (§1.5. ad fin.), the morphological context of **-tí-* (athematic, with mobile inflection) is quite different, in PIE terms, from that of **-tó-* (oxytone thematic). Nevertheless, some benefit may accrue from drawing the comparison, given the preponderance of oxytone zero grades for both formations in late IE. Most directly, we could suppose that if forms like Ved. *rātá-*/Av. *rāta-* show a relatively old substitute full grade (i.e. ReH-tó- ← "RH-tó-"), similar pressures might account for a generalized full grade in *ti*-stems based on roots like **meh₁-* 'measure' (i.e. RéH-ti-/R̥H-téi- generalizes RéH-ti-). In such a way, one could account directly for a form like **méh₁-ti-* (Hom. μῆτις etc., §1.4. above).

3.2. But in addition to CéH-ti- forms, there is evidence for yet another subtype of full-grade *ti*-stems that is attested with some prominence, namely those with root-final *-s* and initial resonant or glide ("Rés-ti-"), a format that is phonotactically reminiscent of the μῆτις type. Thus, a **u̯és-ti-* 'clothing' (abstract → concrete) may be old: Lat. *vestis* 'gar-

[20] In Indic, the process eventually intensifies in a way that is reminiscent of the generalized full grades in Old Iranian: e.g. for *chā-* 'cut', Ved. *chitá-* (SB) competes in late Vedic with *chātá-* (Sutras+); similarly *hā-* 'leave' (RV *jahitá-*, but AB *hāna-*, Class. *hāta-*), *śā-* 'sharpen' (RV *śitá-*, but Ep.+ *śāta-*).

[21] See recently MOTTAUSCH (2000: 34, 38) for a similar conception involving the early replacement of samprasāraṇa zero grades like (gen.) ***m̥dh-éu̯-s* (: **médh-u* 'honey'), ***n̥pt-és* (: **népōt-s* 'grandson, nephew').

ment', Go. *wasti* 'id.', Skt. *vastí-* 'fringe' [gramm. only].[22] Similarly, a **nés-ti-* (to **nes-* 'return') is at least old within Greek, to judge from Myc. *Ne-ti-ja-no /Nesti(i̯)-ānōr/*.[23] Thus, we could again suppose a secondary substitution of full grade roots for the samprasāraṇa zero grades in the weak stems ***us-téi̯-* and ***n̥s-téi̯-*, which might have been disfavored (cf. R̥H-téi̯- above, **§3.1.**). On the other hand, such zero grades certainly exist elsewhere for these roots,[24] so that other explanations should also be considered. For **u̯és-ti-*, we have an acrostatic middle present **u̯és-toi* (Hitt. *wēsta* etc., LIV 633), which, as a category with persistent full grade of the root vis-à-vis ordinary middles, could perhaps have influenced (or otherwise have been regularly correlated with) the full-grade verbal abstract, as already noted for this item by SCHINDLER (1994: 398). For **nés-ti-* (actually **nés-ti°*) we should consider another factor, namely the status of this form as the first member of a *τερψίμβροτος* compound, which returns us to one of the points made at the outset (**§1.4.**), concerning the prominence of CéH-ti- forms precisely in this category. In the remainder of this paper, then, I consider full-grade *ti*-stems from the perspective of original compound formations, beginning with some further observations about CéH-ti- forms and *τερψίμβροτος* compounds.

3.3. The best actual word-equation for a *τερψίμβροτος* compound involves a CéH-ti- form, i.e. the **déh₃-ti-* of RV *dā́tivāra-* (**§1.4.** above) beside the *δωτ(ι)°* implied by the Kurzname Δωτώ (which must stand, as has long been assumed, for Δωτι-κλέης vel sim., cf. KNECHT 1946: 11); cf. also OCS *blago-datĭ* grace, Av. *-dāiti-* 'giving'. Some of the Homeric forms of this sort, such as the *βωτι°* already mentioned

[22] Other material here is less certain, e.g. Hsch. *γεστία*, normally emended to *γέστρα* and compared with Skt. *vástram*. (Toch. B *wastsi*, sometimes also adduced in this connection, is simply the regular infinitive of *wäs-* 'clothe').

[23] The Mycenaean form is directly comparable, mutatis mutandis, to Thess. Νέσσανδρος (cf. GARCÍA RAMÓN 1999b: 526). A further index of the antiquity of **nés-ti-* may be found in the early Gk. **késti-* (< **kéns-ti-*) of Myc. *Ke-sa-do-ro/ra* /Kessandros, -rā/, which may owe its form to the parallel **nés-ti-*compound. See in detail GARCÍA RAMÓN 1992.

[24] Though rarely, and not without controversy in some cases: thus e.g. Ved. *ásta-* n. 'home' (substantivized **-n̥s-tó-*, with accent retraction) seems assured, although the ultimate meaning and analysis of Hom. ἄσμενος and the etymology of Skt. *(sv)astí-* are less clear; for the latter, cf. MEIER-BRÜGGER 1979 (**n̥s-ti-*) vs. the traditional derivation via **h₁es-ti-* (e.g. WATKINS 2000: 24).

(**§1.4.**), remain isolated (qua full-grade *ti*-stems): likewise *βητ(ι)°* in Hom. *βητάρμων* 'dancer' (2x *Od.*)[25] and 'Αναβησίνεως (η 113). But in a case like *Στησ(ι)°* (Στησίχορος etc.),[26] we have parallel CéH-ti-material that is either simplex (Av. *stāiti-* 'resistance' and other meanings[27]) or that appears in other compound forms (ORuss. *po-statĭ* 'portion').

Moreover, as already mentioned, we have a well-attested series of simplex CéH-ti- forms of the type Gk. *μῆτις* (**§1.4.**). Among these, a particularly archaic-looking form is the **dʰéh₁-ti-* implied by Av. *δāiiti-*, Lat. *fēti(ālis)*, Go. *ga-deþs* (among other Gmc. material: OHG *tāt* etc.), Li. *dė́tis*. According to an etymology recently proposed by C. MELCHERT (1997: 34ff.), the same form may also occur in Lydian *ta(a)c-* 'votive tablet/offering', which has interesting consequences for the question of how *ti*-stems otherwise appear in Anatolian. There, of course, plain primary *ti*-stems are virtually unattested, as opposed to productive replacements like the suffix conglomerate seen in Hitt. *-uzzi-* (< **-u-ti-*), and apart from a few alleged plain *ti*-stems in Luvian, which can be (and perhaps should be) interpreted differently.[28] In Hittite, the best proposal for an old primary *ti*-stem appears to be *tukzi-* 'cultivation, breeding', which according to MELCHERT (1999b, see esp. 22f.) may belong with OHG *zuht* 'Zucht' and Lat. *ductim* 'by drawing'. If this is so, it seems highly suggestive that already in Anatolian we would again find the typical Indo-European distribution as described above, i.e., a full-grade CéH-ti- form (if this etymology of Lydian *ta(a)c-* is correct) beside a generalized zero-grade **duk-ti-* in Hitt. *tukzi-*.[29]

[25] KNECHT 1946: 34; but the interpretation of the form is disputed, see FRISK s.v.

[26] Myc. *Ta-t(i)-: Ta-ti-qo-we-u /Stātigʷoweus/* and (probably) *Ta-ta-ke-u /Stātarkʰeus/*, cf. Στήσαρχος (GARCÍA RAMÓN 1985: 201-3); differently on these forms DUNKEL, cf. n. 35 below.

[27] Conspectus in TREMBLAY 1999: 58.

[28] Thus RIEKEN (1999:102n465) is properly skeptical about STARKE's (1985) interpretation of CLuv. *utti-* and *azzašti-* as **hₓgʷ-ti-* 'drinking' and **h₁d-ti-* 'eating', respectively. (See also next fn. on "Hitt." *latti-*.)

[29] This picture is not significantly disrupted even if Hitt. *taišzi-* 'burden' continues a **dʰoh₁-es-dʰh₁-ti-* (RIEKEN 1999: 190), since this is not a primary formation in any case. Rather more suggestive (though still uncertain) is the idea that (Hitt.) *latti-* 'tribal troop' reflects the CLuv. counterpart of Hitt. *tuzzi-* 'army',

There is in any case no lack of CéH-ti- forms beyond those already mentioned, many of which appear relatively isolated within their traditions and may well be archaic. Without attempting an exhaustive presentation (much of the material is readily available in LIEBERT 1949, and for Indo-Iranian see now TREMBLAY 1999: 53ff., 71ff.), it may be helpful to provide some characteristic examples. In Indo-Iranian, we find Ved. *rātí-* 'gift' ~ Av. *rāiti-* 'generosity' (**(H)réh$_1$-ti-*), Ved. *sphātí-* 'fattening' (**spéh$_2$-ti-*[30]), Av. *hāiti-* 'section' < *'binding' (**séh$_2$-ti-*). Beside Ved. *sphātí-*, note also the Gmc. **spōdi-* of OHG *spuot*, OE *spēd* 'success'[31] — evidently an oxytone formation, vs. the barytone type Go. *ga-kumþs* (: Ved. *gáti-*).[32] Some additional examples: **(s)téh$_2$-ti-* (: **(s)teh$_2$(i)-* 'steal') in OIr. *táid,* OCS *tatĭ* 'thief'; **péh$_3$-ti-* (**peh$_3$(i)-* 'drink') in ἀνάπωτις [Pi.], ἄμπωτις [Hdt.+] 'ebb' (cf. also Lat. *pōtiō*, though this can be secondary); **séh$_1$-ti-* (: **seh$_1$-* 'sow') in Germanic (OE *sǣd* 'seed', Go. *mana-seþs* 'mankind', etc.).

3.4. Given the existence of such forms, together with the parallel appearance of some of them in the context of τερψίμβροτος first members (**§§1.4., 3.2., 3.3.**), we can pursue the implications of this convergence by considering next the question of the original accentuation of τερψίμβροτος compounds. Unfortunately, this is not entirely clear: only the Vedic evidence is of any value, and the data are meager and contradictory.[33] Nevertheless, it is surely most reasonable to assume that full-grade first members had accent on the root, as appears directly in

with *latti-* continuing zero-grade **d^{h}h$_1$-tí-* while *tuzzi-* shows the productive replacement with **-u-ti-* (OETTINGER 1994: 313n27, with further references). For *tukzi-*, MELCHERT himself already noted the generalized **dúk-ti-* for the Germanic and Latin comparanda, vs. the expected proterokinetic **déu̯k-ti-/*duk-tei̯-* (1999b: 23n30). As he also points out, one or another of these forms could have been produced independently, given the productivity of **-ti-*. But from the point of view of the conception of *ti*-stems being developed here, the phonotactic and apophonic contrast between **d^{h}éh$_1$-ti-* and **dúk-ti-* (as apparently preserved in Anatolian) is precisely to be expected.

[30] I follow the "Cornell" reconstruction of this root (vs., e.g., LIV **sp^{h}eh$_1$-*), see NUSSBAUM apud JASANOFF 1994: 160nl9.

[31] For the *o*-grade, cf. OE *spōwan* etc. (and further **§3.7.** below).

[32] For traces of a Verner variant **kundi-* see RASMUSSEN 1983[1999]: 87.

[33] See WACKERNAGEL-DEBRUNNER II.1.320.

RV *dātivāra-* and indirectly in *púṣṭigu-* 'cattle-rearing [= PN].[34] It is thus not unthinkable that at least some simplex CéH-ti- forms are decompositional versions of τερψίμβροτος first members, where the full grade was preserved under the accent; then, on the basis of such CéH-ti˚ → CéH-ti- patterns, the CéH-ti- type could have become established (or, to view the matter somewhat differently, could have maintained a foothold into the late IE period) as a full-grade *ti*-stem type in simplicia; i.e.:

**deh$_3$-ti˚* (RV *dāti˚*, δωτ(ι)˚) ~ **déh$_3$-ti-* (Av. *dāiti-*, OCS *blago-datĭ*)
**stéh$_2$-ti˚* (Myc. *Ta-t(i)˚*, Στησι)[35] ~ **stéh$_2$-ti-* (Av. *stāiti-*)

whence

**d^{h}éh$_1$-ti-*, **méh$_1$-ti-*, **spéh$_2$-ti-*, **péh$_3$-ti-*, etc.[36]

And along the same lines, forms like **u̯és-ti-* (Lat. *vestis* etc.) could be accounted for if there were sufficient (and sufficiently old) τερψίμβροτος forms like **nés-ti˚* (Myc. *Ne-ti˚*).[37]

[34] Other forms with zero-grade first members show the oxytonesis of the simplex: *vr̥ṣṭi˚* like simplex *vr̥ṣṭí-* 'rain', etc.

[35] Differently on these forms DUNKEL (1992: 222), according to whom Στησι˚ is a full-grade "si-imperative", while *Ta-t(i)˚* continues a zero-grade *ti*-stem as in στάσις.

[36] In such a case, the oxytonesis of Ved. *rātí-*, *sphātí-* etc. would be secondary in one way or another, which is not a difficult assumption; in addition to pressure from oxytone zero-grade simplicia (cf. n. 34 above), the same result would be predicted in any case by the "laryngeal accent shift" for Indo-Iranian *i*-stems, as proposed by LUBOTSKY (1992).

[37] Likewise (although an actual τερψίμβροτος compound is not involved), it seems worth considering whether **pénk^w-ti-* 'group of five', traditionally considered to reflect an ablauting proterokinetic *ti*-stem abstract (e.g. *EWAia* II.63, BLAŽEK 2000: 108), may ultimately be decompositional: cf. RV *pañktí-rādhas-* 'bestowing a five-fold gift' (I.40.3c), and the fact that all of the probable exponents of this form (in Indo-Iranian, Germanic, Albanian, Slavic, and Umbrian) in fact show full grade (which, to be sure, might easily have been restored on the basis of cardinal **pénk^{w}e*).

3.5. Such a decompositional scenario calls to mind the traditional conception of *ti*-stems as having been primordially compositional (and in systematic opposition to simplex *tu*-stems), as argued in several classic papers by MEILLET and others,[38] though generally with reference to *ti*-stems as second members of compounds, not first members. An updated version of this theory has recently been defended by B. A. OLSEN and J. E. RASMUSSEN (1999, see also RASMUSSEN in this volume). This claim has been vigorously disputed (see in *extenso* LIEBERT 1949: 149ff., with earlier references), and there is no question of attempting to settle the matter here. But whether or not *ti*-stems were *exclusively* compositional in origin, it would be difficult to deny a significant degree of compositional behavior for this formant, beginning from a very early period.[39] As the second member of compounds, **-ti-* is regularly unaccented.[40] If this behavior is old, as it seems to be, it would help to account for the generalization of zero grade of the root and the suffix in simplex *ti*-stems.[41] Viewed in this light, a form like Ved. *matíṣ* (cf. **§1.2.**(iii) above) might have less to do with an original proterokinetic **mán-ti-ns* than with a zero-grade second member of a compound **˙mn̥-ti-* (whence secondarily oxytone **mn̥-tí-*), as is in fact well-attested for this form (Go. *ga-munds,* Li. *at-mintìs*, OCS *pa-mętĭ*, Lat. *āmēns/dēmēns*[42]). Note further that for second-member *ti*-stem zero grades of CeH- roots (i.e. ˙CH-ti-), laryngeal loss would have been phonologically regular, as in Ved. *bhága-tti-* 'generous gift' (with the same treatment as in ˙CH-to-: Ved. *ā́-tta-* to *dā-* 'give', etc.). It is perhaps not surprising, then, that we find at least one ˙CeH-ti- form (RV *havyá-dāti-* 'presenting oblations'), in which the full grade has presumably been restored after the corresponding CeH-ti˙ compound (cf. RV *dā́tivara-*), and/or a simplex CeH-ti- abstract (Ved. ***dāti-*), if

[38] See especially MEILLET 1925; additional references in WACKERNAGEL-DEBRUNNER II.2.634.

[39] As a variant (and this is the position of OLSEN and RASMUSSEN): even if **-ti-* was at first exclusively compositional, simplex **-ti-* is already established by late IE.

[40] WACKERNAGEL-DEBRUNNER II.1.214.

[41] Similarly OLSEN and RASMUSSEN, though with some different assumptions about the suffix.

[42] See ERNOUT-MEILLET s.v. *meminī*.

such a form existed in the Vedic or pre-Vedic period (cf. Av. *dāiti-* and OCS *blago-datĭ*).

Consider, finally, that if we take the more extreme position that *ti*-stems were in origin exclusively compositional, this could mean that the proterokinetic inflection this class displays, while real enough, would have arisen in an entirely secondary way, i.e., as a result of providing a simplex inflectional paradigm to first-member full grades and second-member zero grades. According to OLSEN and RASMUSSEN (1999: 424), the non-canonic "semi-proterokinetic" inflection of *ti*-stems as we actually find them in the late IE type **mn̥-ti-/*mn-téi̯-*, **g^{w}m̥-ti-/*g^{w}m̥-téi̯-*, etc., arose secondarily after the inflection of *tu*-stems (a proterokinetic type originally at home in simplicia, but serving as a derivational base for *ti*-stem compounds, cf. above). This could well have been a factor — cf. **§1.3.** on possible cases of this sort at the einzelsprachlich level — and yet such an explanation would not account for the full-grade CéH-ti- forms, among other *ti*-stem full grades (such as Cés-ti-, and even **mén-ti-*, as in Go. *ana-minds* 'supposition', and perhaps Skt. *manti-* [gramm.], a type of full-grade *ti*-stem structure that is attested with a certain frequency in Indo-Iranian,[43] and similar to the **ĝénh$_1$-ti-* 'birth, generation' that may be reconstructable for PIE,[44] beside other possible forms with that structure.[45])[46]. A final

[43] Ved. *tantí-*, *ránti-* etc., see LIEBERT 1949: 72ff., 108f. This pattern, which can hardly be old (though it may owe its extension to an old prototype), calls to mind the full-grade *ro*-stems to *Tand-* roots (Ved. *átandra-* 'untiring', *candrá-* 'brilliant', etc.), which I discuss elsewhere (2002: 330, 344f., with further references).

[44] Av. *frazainti-*, Gmc. **kindi-* (Go. *kindins* 'governor'), OCS *zętĭ*, etc. (LIEBERT 1949: 107), but not necessarily Gk. γένεσις, which could simply reflect secondarily stressed **ĝń̥h$_1$-ti-*. (For an analysis of **ĝénh$_1$-ti-* in terms of a corresponding acrostatic middle — cf. **ǵués-tor* : **ǵués-ti-* above, **§3.2.** — see SCHINDLER 1994: 399.) Whether Lat. *genti-* (nom. *gēns* 'nation') belongs here remains unclear, given the absence of laryngeal reflex (vs. the zero grade in Lat. *nātiō*); SCHRIJVER may be correct in taking *genti-* as a late creation (1991: 330, following ERNOUT-MEILLET).

[45] Cf. the possible **h$_1$énh$_3$-ti-* of Gk. ἐνοσι-, although the case is ambiguous, as noted recently by JANDA (1999: 187f.). The τερψίμβροτος formation, however, is noteworthy (ἐνοσίχθων etc.).

[46] Even without considering various TéT-ti- forms (e.g. **kéd-ti-* in Lat. *cessim* 'gradually', *necesse* 'unavoidable') or others with still more complex clusters whose full grades can also be secondary (e.g. **rézg-ti-* in Lat. *restis* 'cord', cf. Li. *rezgù* 'plait'), still other full-grade *ti*-stems are met with synchronically, without

complicating factor is that the connections between *to*-participles and *ti*-stems are to some degree very tight (thus explicitly MEILLET 1929: 639), with the result that it is not always easy to interpret the derivational relationships between parallel *ti*-stems and verbal adjectives in **-to-*: cf. Lat. *pōtus/pōtiō* (**§2.5.**), Ved. *rātá-/rātí-* (**§2.7.**), Gk. *˚πᾱ[σ]το-/πᾱσι-*, *˚κτᾱτο-/κτᾱσι˚* (**§2.4.**(iv)), etc.[47]

3.6. The general conclusions to be drawn from this discussion can be summarized briefly. I have attempted to highlight a type of problem in Indo-European word formation that arises directly from the dramatic advances of recent decades in our understanding of accent and ablaut correlations in the Indo-European noun. The general architecture of this theory differs little from one version to another, despite details of terminology and conception associated with individual schools and practitioners (see e.g. MEIER-BRÜGGER 2000: 192): a relatively small set of idealized accent/ablaut models accounts for the vast majority of athematic nominal derivation. Nevertheless, many details of root and suffix alternation do not actually fit the basic parameters of these models, whence traditional assumptions as to secondary generalizations, partly in the histories of particular languages, and partly within PIE itself. Now that a more or less generally agreed-upon set of basic models has been arrived at, it seems worthwhile to examine more closely the sometimes recalcitrant details associated with the presumed generalizations — a type of approach that should perhaps now receive a higher priority

the age of such formations being clear in all cases: isolated examples include Lat. **fū-ti-* < **ĝʰéu̯-ti-* (presupposed by *fū-tilis* 'leaky', cf. also *fūtis* 'vase' and *effūtīre* 'gossip', vs. pres. *fundō* 'pour'), Lat. *vecti-* < **u̯eĝʰ-ti˚* in *vectīgal* 'revenue' (see DUNKEL 2000: 95 on the formation of the compound), RV *váśti-* (hapax, V.79.5c; probably with *vaś-* 'desire', thus **u̯eĥ-ti-*, see *EWAia* II.527 and BENEDETTI 1999: 14n34), and the Germano-Slavic **u̯ék-ti-* of Go. *waihts*, OCS *vešti* 'thing' (IEW 1136). For the structure of **u̯eĝʰ-ti˚*, **u̯éĥ-ti-* and **u̯ék-ti-*, cf. **u̯és-ti-* (**§3.2.**) and the discussion of samprasāraṇa ablaut above. (For **u̯eĝʰ-ti˚*, note further SCHINDLER 1994: 398, correlating Ved. *váhas-*, Av. *vašta-* and similar forms with an acrostatic middle **u̯eĝʰ-to*, cf. **§3.2.** and fn. 44 above.) Cf. also Myc. *E-ne-ke-se-u /Enekseus/*, which points directly to a τερψίμβροτος form **Enek-si˚* < **hₓneĥ-ti˚*, see GARCÍA RAMÓN 1999c: 69 and (on the hypocoristic formation in general) MORPURGO DAVIES 1999. (The apparent **mét-ti-* of Lat. *messis* 'harvest' could have been reshaped after *metō* 'mow'.)

[47] Thus here also perhaps **τᾱtó-* (if this is indeed the basis of τητάομαι, cf. n. 14 above)/OIr. *táid*, OCS *tatь* (**§3.3.**).

than heretofore.[48] In some cases, the theory itself may need to be adjusted, and I have tried to suggest one such adjustment or enhancement that strikes me as theoretically quite interesting, namely the idea that a range of external factors (such as root-structure constraints, and the features of certain compound formations) may be responsible, at least in part, for the special nature — and possibly even the very existence — of an irregular inflectional pattern, in this instance the skewed phonotactic distribution of *ti*-stems and *ti*-stem alternations as they must actually be reconstructed for late PIE.

3.7. In view of the above comments, it may be fitting to conclude by signaling a problem that deserves further investigation, namely *ti*-stems with *o*-grade of the root.[49] In some such cases, a corresponding *o*-grade present may provide a more or less direct source for the formation: in addition to Gmc. **spōdi-*: **spō(w)an* (**§3.3.** and n. 31 above), note e.g. Gmc. **hait-ti-* (Eng. *hest, behest* etc.): **haitan* (Go. *haitan* 'name, call'), or the apparent **u̯old-ti-* of OCS *vlastь* 'power, dominion' (cf. OCS *vladǫ/vlasti*, Go. *waldan* 'overcome').[50] In a similar way, some forms are likely to be einzelsprachlich creations based on other *o*-grade forms in the language: thus, there is little chance that Gk. πτῶσις 'fall' (cf. also πτῶμα 'id.') continues an old **ptohₓ-ti-*, rather than a secondarily-derived abstract to the (unreduplicated) stem of pf. πέπτωκα (itself with a secondary vocalism, see LIV 429f.).

There are, however, several candidates with a much more isolated cast, and which may thus point in the direction of an old category of R(ó)-ti- forms, e.g.:

(i) **gʰós-ti-* 'stranger/guest' as simplex (Lat. *hostis*, Go. *gasts*, OCS *gostь*) and — at least with some probability — in a compound **gʰosti-pot-* (Lat. *hospes*, OCS *gospodь*) appears to be reconstructable for some stage of the protolanguage;

[48] In addition to various papers presented at this conference, see now also MOTTAUSCH 2000 for an interesting attempt to integrate accent/ablaut theory, including certain assumptions about accented *o*-grades, with the often puzzling data of such *o*-grades themselves.

[49] Such forms are not treated by MOTTAUSCH (op. cit., cf. preceding footnote).

[50] Thus also various *e*-grade *ti*-stem in Gmc.: **gʰébʰ-ti-* 'gift' (ON *gipt*, cf. *gefa* 'give'), **gʷét-ti-* 'bequest' (OE *-cwis*, cf. *cwepan* 'speak'), **u̯éĝʰ-ti-* 'weight' (OE *wight*, cf. *wegan* 'carry, weigh'), etc.

(ii) entirely isolated within their tradition are the **món-ti-* 'projection' of Lat. *mōns* 'mountain' (cf. *ēmināre* 'jut, project', *mentum* 'chin') and the **dʰon[h₂]-ti-* 'flowing' of Lat. *fōns* 'fountain' (cf. Ved. *dhánvati* 'runs, streams', causat. *dhanaya-*);

(iii) the RV hapax *jígarti-* 'swallower' (or the like), attested in an intriguing phrase with the appearance of a *figura etymologica* (*jígartim índro apajárgurāṇaḥ*, V.29.4c),[51] raises a series of morphological difficulties, as discussed in detail by BENEDETTI (1999: 14-16, with nn. 34-36): the reduplication is unexplained, and, given the lack of palatalization in the root, we appear to have an *o*-grade **(-)gʷor-ti-*, which also lacks the laryngeal of **gʷerh₃-* 'swallow' (LIV 189, Ved. *giráti* etc.). For the latter feature, BENEDETTI rightly rejects a solution via laryngeal loss in composition, but she does not note that an *o*-grade sequence of this sort would regularly have undergone a "Saussure's Law" laryngeal loss in any case, however this phenomenon is to be interpreted at the PIE level. While BENEDETTI prefers to view the form as a nonce creation (especially in view of its phraseological context), she does not exclude the possibility that the form is a lexical archaism; in such a case, we would have another *o*-grade *ti*-stem **gʷór[h₃]-ti-*.[52]

On the principle that the *o*-grade of such forms must be explained away secondarily, one may of course posit a secondary derivation on the basis of some *o*-grade source (cf. Gmc. **spōdi- : *spō(w)an* etc. above) — whence, for example, various claims (impossible to substantiate) that Lat. *fōns* and *mōns* are based on *o*-grade root nouns (ERNOUT-MEILLET s.vv.), or that they both reflect contaminations of original *ti*-stems with thematic *o*-grade nouns in **-to-*, i.e. (e.g.) **dʰn̥-tí-* (or pre-Lat. **fenti-*) X **dʰón-to-* (pre-Lat. **fonto-*) → Lat. *fonti-* (thus IEW 249, 726). If, however, such *o*-grade *ti*-stems continue an archaic primary formation, one may be forced to consider the existence of an acrostatic source: it has now become clear that some proterokinetic (or "quasi-proterokinetic") forms have arisen secondarily from earlier acrostatics, as in **ĝón-u-/*ĝén-u-* 'knee'→ **ĝón-u/*ĝn-éu̯-*, and

[51] Extended discussion of the passage (and of *jígarti-*) in BENEDETTI 1999.

[52] Since the stem βρω- for this root has been generalized in Greek, βρῶσις 'food' (concretized abstract/action noun, as if from **gʷr̥h₃-tí-*) need not be particularly old.

several other possible cases of this sort.[53] Whether this is in fact the best way of accounting for forms like **gʰós-ti-* (and **gʰós-ti•*, cf. **déh₃-ti•* etc. above) and similar forms remains a question for the future.[54]

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[53] See recently STÜBER 1997 on two possible *n*-stem examples, and RIEKEN 1999: 190 (cf. n. 29 above) on Hitt. *taiš-* < (acrostatic *s*-stem) **dʰoh₁-(e)s-*. Somewhat differently on *mōns* (vis-à-vis Av. *maiti-* 'summit'): TREMBLAY 1999: 132.

[54] This paper reflects an expansion of what began as a sub-section (on Myc. *E-ti•* and *O-ti•*, Hom. 'Ορτι•) of a study of "Rix's Law" in Greek, originally presented at the 19th East Coast Indo-European Conference (University of Georgia, June 2000). (A preliminary version of that paper is in preparation as VINE 2003.) For helpful comments and suggestions, I am deeply grateful to the participants of the conference (for the *ti*-stem material: especially Craig Melchert and Alan Nussbaum), as well as to the participants of the Copenhagen meeting on Indo-European word formation.

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